

Make This Joystick-Controlled Industrial Automation System

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This project can be used to control up to four industrial electrical appliances with the help of a joystick and an Arduino Nano board. The block diagram of the joystick-controlled industrial automation system is shown in Fig. 1.

The circuit diagram of the joystick-controlled industrial automation system is shown in Fig. 2. It is built around Arduino Nano (Board1), 5V regulator 7805 (IC1), four npn transistors (T1 through T4), joystick module (JM1), transformer (X1), four 12V single-changeover relays (RL1 through RL4) and a few other components.

Joystick module. Joysticks are available in different shapes and sizes. A typical joystick module is shown in Fig. 3. This joystick module provides analogue voltage outputs. Output voltage changes according to direction of the shaft. You can get direction of shaft movement by interpreting voltage changes using a microcontroller (MCU).

This joystick module has two axes: x and y. Each axis is controlled by shaft of the potentiometer mounted on of the joystick module. Potentiometer outputs VRx and VRy at connector CON1 are variable points. VRx and VRy act as the voltage divider.

When joystick shaft is in normal position, it is in standby mode. When it is moved along horizontal axis (right and left), voltage at VRx pin changes. Similarly, when it is moved along vertical axis (up and down), voltage at VRy pin changes. So, we have four different output signals for four directions of the joystick, which are connected to two analogue-to-digital converter (ADC) input pins A0 and A1 of Arduino Nano.

When the shaft is moved, voltage on each pin goes high or low depending on direction. The joystick module also comes with one push-button switch for interruption purpose

(this function is not used here). The joystick module is interfaced with

Arduino Nano, which comes with an inbuilt ADC, as shown in Fig. 2.

Arduino Nano drives four relays (RL1 through RL4) through its pins D2, D3, D4 and D5.

Power supply. Regulated power supply is designed with a full-wave rectifier circuit. It consists of D1 and D2 (1N4007) rectifier diodes, capacitors C1 and C2, IC1 7805 and 230V AC primary to 12V-0-12V, 750mA stepdown trans-

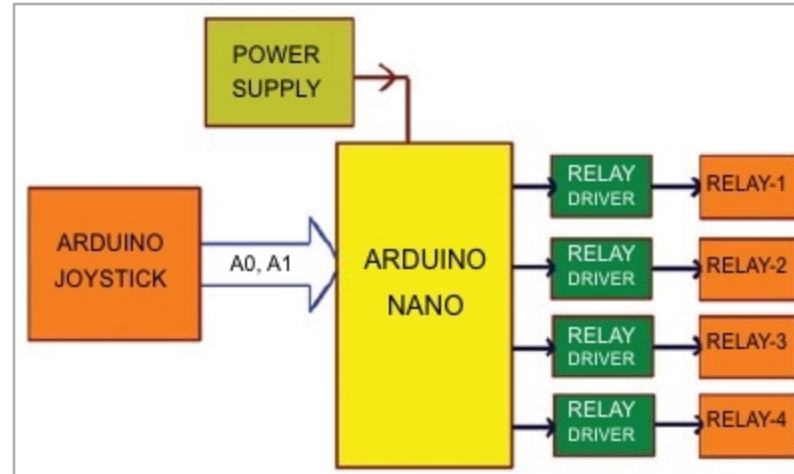


Fig. 1: Block diagram of the joystick-controlled industrial automation system

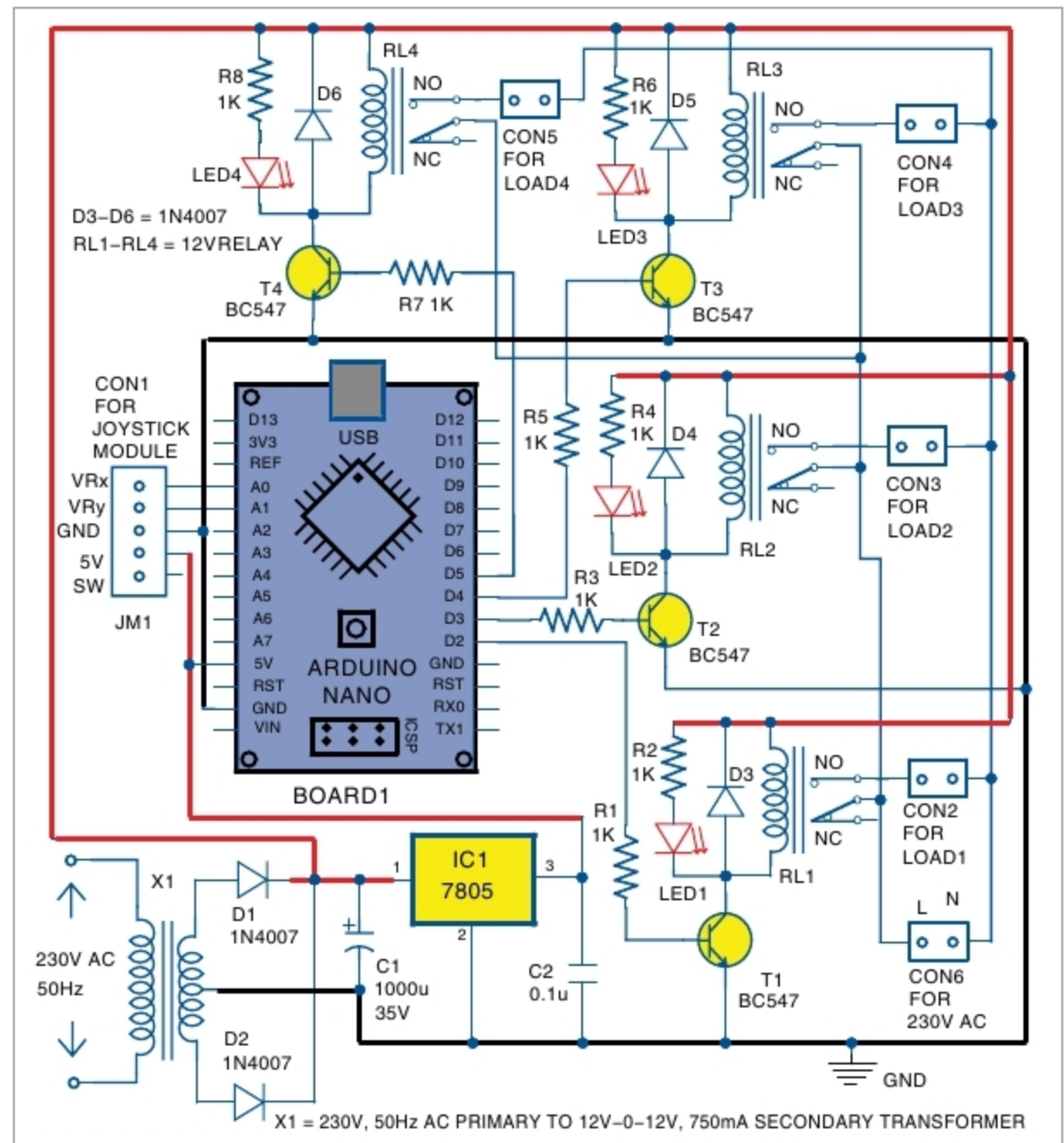


Fig. 2: Circuit diagram of the joystick-controlled industrial automation system